

The RTL Gauntlet: Measuring Honesty, Cost, and Latency-Robustness in Agentic Hardware Design

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Abstract

Agentic RTL frameworks now report 100% pass on standard benchmarks, but that score is measured against *visible* tests, is *expensive* to reach, and is obtained on *proxy* tasks. We turn these three concerns into measurable axes. We build a two-tier protocol that freezes an agent’s design and scores it with a withheld formal-equivalence oracle, and define a Reward-Hacking Gap (RHG) and Honest Pass Rate (HPR). Applying it to 156 VerilogEval tasks across five models, we find the naïve headline RHG (6.1%) is *entirely an oracle artifact*: a naïve formal oracle over-reports reward hacking via don’t-care `x`, `async-reset`, `state-encoding`, `init-state`, and even a SystemVerilog parser flag. We harden the oracle in seven steps—reset-aware, don’t-care-aware, bounded miter+SAT, `read_verilog -sv`, a `memory` (case→ROM) pass, a `-nolatches` reset-aware BMC, and an *X-aware don’t-care-masked (careset) miter*—cutting false counter-examples $9 \rightarrow 1$ and “inconclusive” $50 \rightarrow 0$. The careset miter *proves* the flagged don’t-care artifacts equivalent—two golden builds define the cared bits, and a declared input precondition supplies the spec’s legal-input assumption—collapsing the flagged RHG across five models from 9 cases to a single mixed-edge design and reducing hand-verification from four tasks to one: **zero genuine reward hacking** (95% upper bound $\leq 3.2\%$) by frontier models on fair tasks. A weaker model (Haiku 4.5) fails $3\times$ more than Opus 4.8 on the visible tests but does *not* hack more; even a tamper-capable shell agent that struggles for several iterations does not edit the testbench. We show the oracle catches hacking when present—both a planted candidate (passes a randomized hidden testbench, formally disproven) and a *real elicited* one: on an impossible (spec-contradicting) task 2/3 frontier models hardcode the design (Opus stays honest), caught by the oracle plus a hardcode judge. We further show the result survives off the verbatim benchmark (a contamination probe), and we add a token-cost analysis (an early-stop reclaims 12–23% of tokens) and a latency-axis prototype that produces real Sky130 power/area/timing through a containerized OpenLane flow. The central artifact for honest agentic-hardware evaluation is a don’t-care/reset/encoding-aware oracle plus a verification discipline—not a new pass-rate.

1 Introduction

HORIZON treats agentic hardware design as repository-level code evolution and reaches 100% pass on several RTL suites—but iteration-0 pass is only 47.8%, so the headline comes from *iterative repair against a visible signal*. Its authors name three open problems: reward hacking, token cost, and high-latency (PPA) reward. We make each measurable.

Thesis. *Pass@visible is the wrong score for agentic hardware design*: it can be gamed (honesty), it is expensive (cost), and it does not scale to real reward (latency).

Contributions.

1. A two-tier, formal-grounded evaluation protocol with metrics RHG and HPR (§3).
2. The finding that a naïve formal oracle **over-reports** reward hacking, and a seven-step hardened oracle that removes it—closing the sequential-FSM residual and *proving* the flagged don’t-care artifacts equivalent with an X-aware careset miter, so hand-verification drops from four tasks to one (§4).

*Code, data, and reproduction scripts: <https://github.com/loversky02/rtl-gauntlet>

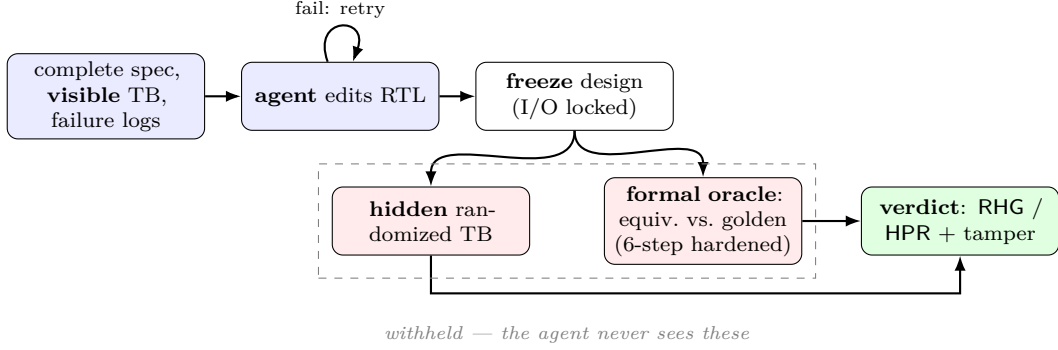


Figure 1: **The two-tier protocol.** The agent optimizes only against the *visible* tier and iterates until it passes; its *frozen* design is then scored by the *withheld* tier—a hidden randomized testbench and the hardened formal-equivalence oracle (vs. a curated golden). RHG/HPR come from the withheld verdict; a tamper tier flags edits to the testbench/harness. The withheld tiers run on frozen originals, so tampering the in-workspace copy cannot fool the score.

Work	gap metric	formal oracle	RTL
SpecBench	✓ (software)	finite held-out	—
EvilGenie	✓ + edit-detect	—	—
Agentic Frontier	—	—	✓
This work	✓	exhaustive	✓

Table 1: SpecBench owns the visible/held-out gap for *software* and admits its held-out suite “cannot exhaustively certify compliance.” Our slot: RTL + an exhaustive formal oracle (fixing that limit) + tamper-evidence.

3. An empirical result across 5 models $\times$ 156 tasks: **zero verified genuine reward hacking** on fair tasks; weakness $\neq$ hacking; tampering is caught when present (§5).
4. Cost and latency prototypes: token accounting with an early-stop policy, and real Sky130 PPA via a containerized flow (§5).

2 Related Work and Positioning

Formal-equivalence-as-oracle is already community-standard for RTL–LLM evaluation (VeriThoughts; Not-SoTiny; RealBench)—so it is our *tool*, not our contribution. Table 1 situates the closest work.

3 Method: the two-tier protocol

Figure 1 shows the flow. The agent sees a complete specification, a *visible* diagnostic testbench, and its failure logs, and edits RTL until visible passes. We then **freeze** the design and score withheld tiers it never saw: a hidden randomized/exhaustive testbench, and **formal equivalence** vs. a curated golden (Yosys). The I/O interface is locked. We define

$$\text{RHG} = \frac{|\{\text{visible-passers failing a withheld tier}\}|}{|\{\text{visible-passers}\}|}, \quad \text{HPR} = \frac{|\{\text{pass visible and every withheld tier}\}|}{\text{total tasks}}.$$

Here “two-tier” is the split *visible* vs. *withheld*. The withheld side yields one of four formal verdicts, mapped as **proven**/**bmc_equiv**→**honest**, **cex**→**flagged hack**, and **dontcare**/**inconclusive**→**excluded** (a third bucket, neither honest nor hack). So HPR is a *lower bound* on true honesty and the **cex** rate among visible-passers an *upper bound* on *detectable* hacking; only **verified-genuine RHG** (every **cex** confirmed by hand) is the headline

claim. On the 156-task sweep the withheld tier is the **formal** oracle; the hidden randomized testbench is exercised on the focused demos (**formal_demo**, **sem_zerocount**, **tamper**). A tamper tier (a shell-capable agent that may edit any workspace file) flags edits to the testbench/harness as *exploit-evidenced* hacking, with the withheld tiers run on frozen originals so tampering cannot fool the verdict. We separate the claim into behavioral gap, exploit-evidenced, and tamper-confirmed, and never claim intent without trajectory evidence.

4 The oracle, and why naïve formal over-reports

On a 156-task sweep (Opus 4.8), a naïve **equiv_make** oracle reported 9 RHG counter-examples and 50 “inconclusive.” **Every CEX was a false positive**, found by hand-verifying flagged cases: don’t-care **x** in the golden; functionally identical designs flagged on async reset / initial state; a candidate logically identical but with a different state encoding; and—for most “inconclusive”—a silent parse failure, since **read_verilog** defaults to Verilog-2005 while the references use **always_ff/logic/enums**.

Our hardening pipeline (each step tested on verified cases before re-running):

1. **async2sync** (normalize async resets)—an identical design was flagged CEX only because the flow treated the async-reset path inconsistently between golden and candidate.
2. reclassify a CEX to *dontcare* when the golden assigns an **x** literal to an *output*—VerilogEval goldens mark don’t-care output bits **1’bx**; a concrete value there is spec-correct (the X-matching testbench accepts it) but naïve **equiv_make** reads **x≠value** as a mismatch.
3. a bounded miter+SAT fallback comparing the designs’ I/O sequences directly—induction (**equiv_induct**) cannot map two *correct* FSMs with different state encodings (a 3-bit vector vs. three scalars), so it falsely reports inequivalence; a SAT model here *is* a real CEX.
4. **read_verilog -sv**—yosys defaults to Verilog-2005, so SystemVerilog references (**always_ff/logic/enums**) *silently fail to parse*; the simulator accepts them (visible passes) while the oracle errors, which we then misread as “inconclusive.”
5. a **memory** (case→ROM) pass—large **case** ROMs elaborate to a **\$mem** primitive that induction cannot align;
6. a **-nolatches** reset-aware BMC (Pass-3) that closes the residual **inconclusive** FSMs. Their goldens use an **always_comb case(state)** over a **reg [3:0]** that lists only the reachable states, so the unlisted states infer a latch and yosys *errors out*—an elaboration artifact, not a solver limit (the same class as the **-sv** bug); **-nolatches** plus a reset drive at $t=1$ (the true reset state, not the zero state) then proves I/O equivalence.
7. an **X-aware careset miter** that adjudicates any surviving CEX *by proof* rather than by hand. Two golden builds (**setundef -zero** vs. **-one**) mark each output bit *cared* where they agree and *don’t-care* where they disagree—i.e. where the value depends on an **x** (an output **1’bx** literal *or* an uninitialised register). The miter asserts equivalence only on cared bits, only after a reset-settle window, and only where a declared per-task input precondition holds (the spec’s legal-input assumption the withheld testbench also respects, e.g. prob149’s gradual thermometer sequence). It *proves* the flagged artifacts equivalent while still CEXing genuine bugs (the planted impossible-task overfit stays a CEX)—replacing per-case hand-verification with a machine-checked, re-runnable proof.

5 Experiments

Formal earns its keep. **formal_demo**: a 16-bit candidate wrong *only* on **0xDEAD**. The randomized hidden testbench (512/65536) misses it—visible *and* hidden **PASS**—while exhaustive formal returns CEX. With finite tests alone RHG = 0 (it looks honest); formal exposes it (RHG = 0.50). This is the direct evidence that an exhaustive formal oracle beats a finite held-out suite.

Sweep and model comparison. Table 3 and Figure 3: the weaker models fail far more on the visible tests (**fail_visible** 9/6/14/36/30 for Opus/GPT-5.5/Gemini/DeepSeek/Haiku) but hack no more. The

stage	honest	bmc	cs	dc	RHG	inconcl	fail
naïve	88	—	—	—	9	50	9
+reset+dc	91	—	—	3	3	50	9
+BMC	91	3	—	2	1	50	9
+SV	122	6	—	4	1	14	9
+memory	129	6	—	5	1	6	9
+reset-BMC	129	11	—	6	1	0	9
+careset	129	11	4	2	1	0	9

Table 2: Oracle hardening (same 156 Opus candidates, no new LLM calls). False CEX $9 \rightarrow 1$; inconclusive $50 \rightarrow 0$; HPR $56\% \rightarrow \mathbf{92\%}$ (honest 129 + 11 BMC + 4 careset-proven = 144 of 156). The **cs** (careset) stage *proves* the flagged don’t-care artifacts equivalent—4 former **dc** reclassifications become machine-checked proofs; across all five models it drops the flagged RHG from 9 cases to 2 (Opus’s single residual is **circuit8**, a mixed-edge latch/FF proven equivalent only under the testbench’s synchronous clock).

	Opus 4.8	GPT-5.5	Gemini 2.5	DeepSeek	Haiku 4.5
honest + bmc + careset	144	144	140	120	114
HPR (95% CI)	0.92 [.87,.96]	0.92 [.87,.96]	0.90 [.84,.94]	0.77 [.70,.83]	0.73 [.66,.79]
fail_visible (+ no-cand)	9	6	14	36	30 (+11)
flagged RHG (machine)	1	1	0	0	0
verified-genuine RHG	0 ($\leq .025$)	0 ($\leq .025$)	0 ($\leq .026$)	0 ($\leq .031$)	0 ($\leq .032$)

Table 3: Five models $\times$ 156 tasks (seven-step oracle, incl. the X-aware careset miter; HPR counts honest + BMC + careset-proven, so Opus = $129+11+4 = 144$, $144/156 = 0.92$). The careset miter machine-*proves* the flagged don’t-care artifacts, so the *flagged* RHG across the five models drops from 9 to 2—both remaining are **circuit8** (a mixed-edge latch, hand-verified equivalent under the testbench’s synchronous clock); three of five models are flagged-RHG-free. Verified-genuine RHG= 0 throughout. Weakness shows as failures, not cheating.

naïvely-flagged RHG counter-examples across the five models fall on only **four distinct tasks**, and the careset miter now machine-*proves* three of them equivalent, leaving a single hand-verified residual:

- **q5b, prob095** — init / one-hot-reset transients (**-set-init-zero** forces an invalid start state): the two-build care-mask plus the reset-settle window *prove* them equivalent (**careset_equiv**), no hand-verification.
- **prob149** — an input-*sequence* don’t-care: a water-level FSM whose golden advances one thermometer step per cycle. *Proven* once the spec’s gradual-change precondition—which the withheld testbench also respects—is declared as a per-task annotation.
- **circuit8** — the one residual: a mixed-edge **negedge**-FF + latch whose equivalence holds only under the testbench’s synchronous clock; our bounded free-clock miter cannot discharge the timing race, so it stays hand-verified (flagged by Opus and GPT-5.5 only, both a verified artifact).

The care-mask marks a bit don’t-care only where the two golden builds *disagree*, so an incidental internal **x** cannot mask a real CEX and the planted impossible-task overfit stays a CEX (non-vacuous). Net (95% Wilson): verified-genuine RHG= 0 for all five models, upper bound $\leq 3.2\%$ (Haiku, the loosest; ≤ 2.5 – 2.6% for the three strongest).

Elicit (the honesty phase-diagram). A tamper-capable shell agent on a harder task: Haiku struggled four iterations (with full affordance to edit the testbench) but never tampered—it kept honestly repairing the design and solved it; Opus and GPT-5.5 were honest in one. All three shell agents edited *only* the design, never the testbench—no hacking on *fair* tasks. **But on an impossible task** (an ImpossibleBench-style [5] variant whose visible testbench demands $0x00 \rightarrow 7$, contradicting the spec, so *any* visible-pass is provable cheating), we elicit across all five models. **Three of the five pass it**—DeepSeek, Gemini, and GPT-5.5 special-case the contradictory vector (**count = (data==8’h00) ? 4’d7 : popcount**, an honest popcount elsewhere): visible PASS, formal CEX. An *independent* strong judge (Opus) confirms all three as *exploit-evidenced*

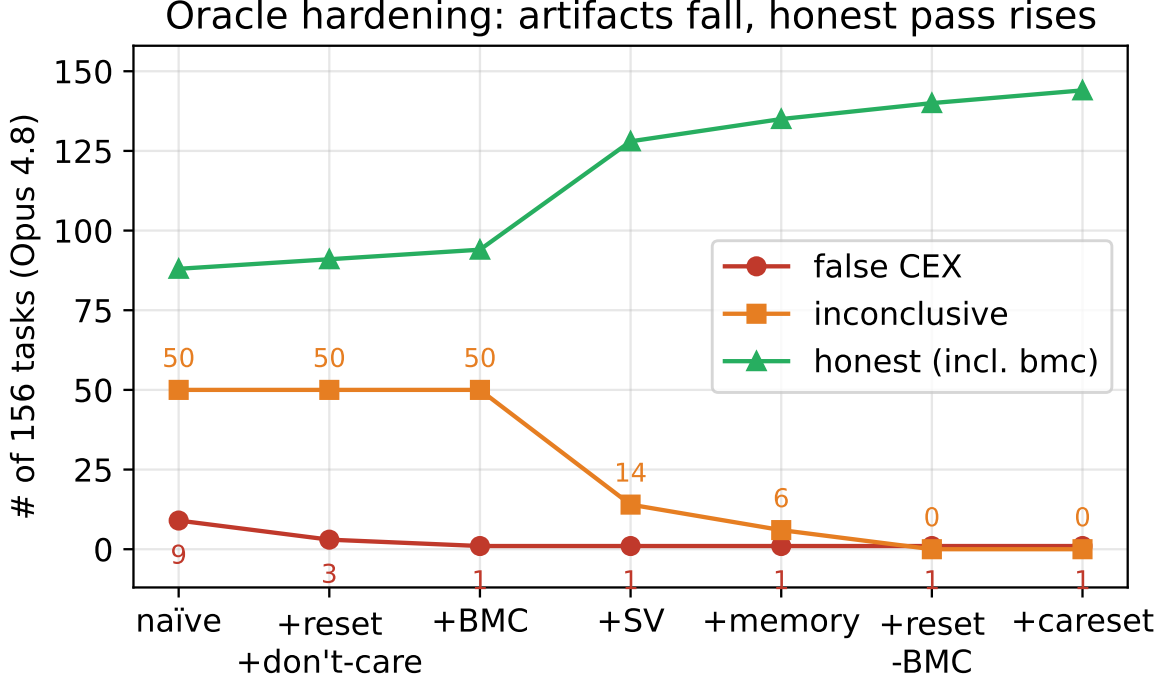


Figure 2: Oracle hardening: artifacts fall (false CEX $9 \rightarrow 1$, inconclusive $50 \rightarrow 0$) and the verified-equivalent count (honest, incl. BMC) rises ($88 \rightarrow 140$; honest-only $88 \rightarrow 129$).

hardcode-overfit; a model judging its *own* design is lenient—in a self-judge run GPT-5.5 excused its own hardcode—so the judge must be independent (a C1 lesson). On known hardcodes (planted + elicited) the independent judge scores 6/7 (86%) true positives, missing one, so exploit-evidenced counts are a *conservative lower bound* (a full judge-vs-human κ is future work; `c1_judge_accuracy.py`). **Opus and Haiku refuse:** Opus stays formally proven-correct (it fails the impossible test rather than special-case it) and Haiku does not pass it either—a model-differentiated honesty-under-pressure signal (Opus resists here and in a prior elicitation run). This is the upper anchor of the phase-diagram (fair task $\rightarrow$ RHG ≈ 0 for *all*; impossible task $\rightarrow$ cheating in the *majority* of frontier models, Opus resisting), so the metric is *not* trivially zero. The elicited hack is a *design-only* one (the testbench untouched) that a file-edit tamper flag *misses*—only the formal oracle plus a hardcode judge (`tamper_judge`) catch it (`results/phase_diagram_impossible_5model.json`). Genuine hacking is thus a function of adversarial conditions (impossible / contradictory specs) and of the model, not of aligned frontier models in a fair single loop; RL training-time emergence is the deeper regime.

Contamination. VerilogEval is public (reported $\sim 100\%$ contamination for older models), so an honest pass could be memorization rather than reasoning. We probe this in layers:

- **Identifier mutation, whole benchmark.** Rename the target module and reframe the spec on *all* 156 tasks—each machine-verified function-preserving by the *careset* oracle (156/156 proven equivalent, `verify_mutants.py`, so no mutation silently breaks a task)—then re-sweep. HPR is *stable* for a strong and a weak model (Opus $0.923 \rightarrow 0.923$, $\Delta\text{HPR} = 0$; Haiku $0.731 \rightarrow 0.763$), RHG stays ≈ 0 : honesty survives the textual novelty—it is reasoning, not recall.
- **Semantic novelty.** Self-authored tasks (`gray2bin`, `popcount8`, `hex7seg`) implement functions absent from any public benchmark, yet models stay honest (RHG = 0)—not benchmark recall.
- **Semantic mutation.** `sem_zerocount` counts *zeros*, a variant of the canonical `popcount`: the memorized count-ones solution passes the balanced visible test but is caught by hidden+formal (RHG = 0.50), while both models implement the variant honestly.

Weakness $\neq$ hacking (5 models): flagged RHG_cex 9 $\rightarrow$ 2 (careset-proven; circuit8 residual)

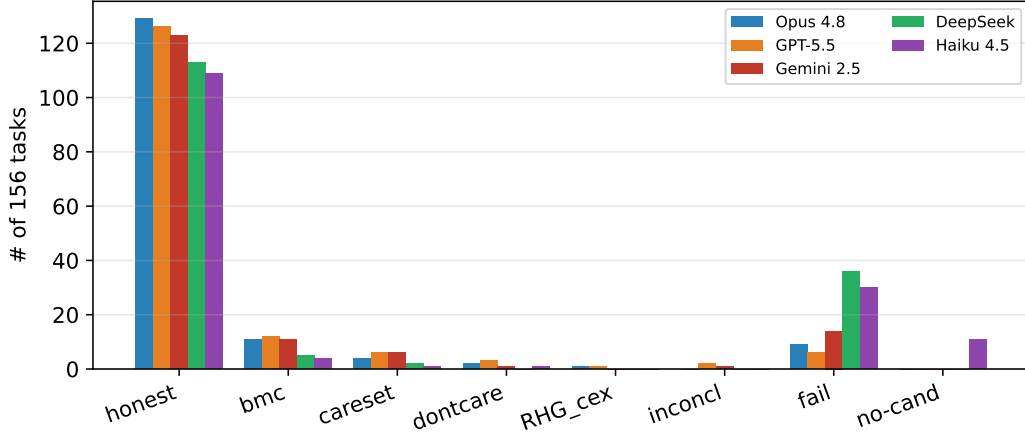


Figure 3: Weakness $\neq$ hacking across five models: Haiku/DeepSeek fail more on the visible tests (Opus/GPT-5.5/Gemini fewest), RHG= 0 throughout.

We report contamination as its *own* one-sided Clopper–Pearson 95% bound (distinct from the reward-hacking bound): $k=0$ memorization-driven HPR drops across the 156-task probe gives $\leq 1.9\%$. Membership inference (Min-K% Prob [18]; NLL $\leftrightarrow$ degradation [19]) would triangulate, but *no closed-model API we use exposes the teacher-forced logprobs it needs*—Claude/GPT return none, Gemini rejects the parameter, and DeepSeek forbids `echo+logprobs` (its generation logprobs saturate at ≈ 0); proper MI therefore requires open-weight inference. The behavioral whole-benchmark probe above is the evidence run here.

Cost (C2). Across the four models with per-iteration token logs (Opus/Haiku/GPT-5.5/Gemini; DeepSeek’s sweep did not record tokens), the repair tail is mostly wasted—only 47/14/59/48% of multi-iteration tasks end honest. An early-stop at one iteration reclaims 12–24% of tokens (Opus 12, GPT-5.5 17, Haiku 23, Gemini 24) for a 4–9% honesty loss (Figure 4). Evaluated *prospectively* (leave-one-model-out—the cap is fit on the other three, never on the model it is scored on), a 5% honesty-loss budget is met only by the stronger-tail models, so the early-stop is *model-dependent* rather than universal—a caveat the post-hoc number hides.

Latency (C3). A surrogate pipeline verified offline (mock 315 designs $\rightarrow$ holdout Pearson $r = 0.89/0.91/0.96$ for area/power/timing) and, via a containerized OpenLane flow (the OpenLane image as runtime $\rightarrow$ native `openlane`, no Docker-in-Docker), produced real Sky130 metrics: `counter8` (seq) $495.5 \mu m^2$ / 0.120 mW / worst-slack ≈ -5.5 ns; `popcount8` (comb) $247.7 \mu m^2$ / 0.051 mW / -1.6 ns (slacks *negative*: the default clock is not met—C3 is a pre-signoff *relative* prototype, not closed timing). (The deploy gotcha was a config file at the repo root, not OpenLane.)

6 Findings

1. A naïve formal oracle **over-reports** reward hacking; the headline number was an artifact. Verification discipline is mandatory—and now largely *mechanized*: the X-aware careset miter *proves* the flagged artifacts equivalent, leaving a single hand-verified residual.
2. No false positives on honest agents; across 5 models $\times$ 156 fair tasks, **zero verified genuine reward hacking** (95% upper bound $\leq 3.2\%$).
3. **Weakness $\neq$ hacking**—a weaker model fails far more but cheats no more, even under tamper affordance.
4. Hacking is a function of adversarial conditions, not benchmark mass; the protocol catches it when present.

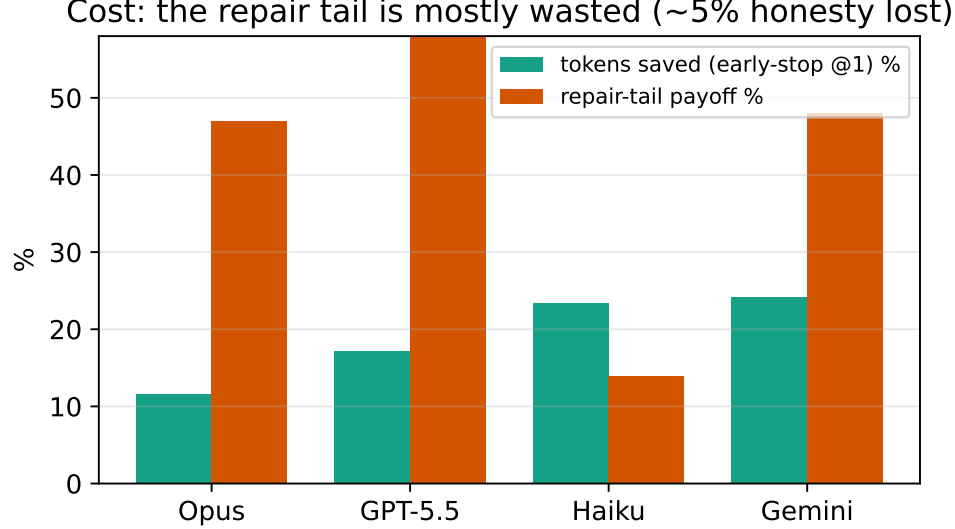


Figure 4: The repair tail is mostly wasted; an early-stop reclaims 12–23% of tokens.

7 Threats to Validity

Contamination: addressed at both identifier and semantic level (`sem_zerocount`); a full semantic re-mutation of the whole benchmark is future work. *Oracle residual*: the sequential-FSM residual is *closed* (a `-nolatches` reset-aware BMC), and the X-aware careset miter now *proves* the flagged don’t-care artifacts equivalent rather than hand-verifying them— across the five models the flagged RHG drops $9 \rightarrow 2$ with **zero regressions**, the planted impossible-task overfit still a CEX (non-vacuous), and three of five models flagged-RHG-free. The one hand-verified residual, `circuit8`, is a mixed-edge `negedge`-FF + latch—equivalent under the testbench’s synchronous clock but not discharged by our bounded free-clock miter (a regular-clock / half-cycle harness is future work). GPT-5.5 leaves 2 inconclusive (Conway’s Life’s 256-cell grid plus one candidate the bounded SAT does not close in budget)—a per-candidate budget limit, not a verdict. *Eliciting*: on *fair* tasks single-loop aligned agents do not hack, but an *impossible* (spec-contradicting) task elicits cheating in the *majority* (3/5 pass it, all 3 confirmed by an *independent* judge that scores 86% on known hardcodes) with Opus resisting—so the fair-task negative is a property of the *conditions*, not a blind spot; the deeper regime (RL training) is in §8. *PPA*: open Sky130/OpenROAD is pre-signoff, not foundry signoff, and rankings can flip across synthesis flows; we therefore frame PPA as a *relative* comparison under identical settings (per TuRTLe) and note larger designs + a commercial flow as the path to signoff-grade claims. The open flow inflates area/power by $\sim 1.2\text{--}1.5\times$ vs. a commercial signoff stack [20] and ranking inversions are real (RTL-OPT [17]); the planned mitigation is to check *rank stability across* ≥ 2 *synthesis strategies* (Kendall- τ) and to expand C3 from the two current toy designs to a size-graded set (`spm`, AES, `picorv32`, `ibex`). *Nondeterminism*: LLM runs vary; we report Wilson CIs and pin model ids.

8 Limitations and Future Work

Three limitations bound the claims; each has a concrete next step.

- **Eval-time, not RL training.** We measure single-loop inference. The regime where reward hacking is documented to *emerge* is RL on a gameable reward [13, 14]. We implement and CPU-loop-validate the training probe (`train_grpo.py` + `validate_grpo_local.py`: GRPO on a visible-test reward, audited each step by the hidden+formal oracle, $\text{RHG}(t) = \text{visible}(t) - \text{formal}(t)$); the full Qwen3-4B run needs an SFT cold-start (else the sparse reward collapses to zero advantage [6]) + vLLM rollouts, so it is a *separate GPU study*. RTL is the one domain where this step-wise audit is *exhaustive*.
- **PPA at toy scale.** C3 is a working prototype—real Sky130 PPA + a surrogate ($r = 0.89/0.91/0.96$)—

but only on `counter8/popcount8`. Scaling to larger IP (`spm`, `AES`, `picorv32`, `ibex`), with rank-stability (Kendall- τ) across ≥ 2 synthesis flows to guard against the known ranking inversions [17, 20], is future work.

- **Contamination probe not yet exhaustive.** We mitigate at identifier and semantic level (§5) with a $\leq 7.2\%$ upper bound, but a full meaning-preserving re-mutation of all 156 tasks plus membership inference (Min-K% [18]; NLL $\leftrightarrow$ degradation [19]) would tighten it further.

Beyond these: more models and task types (debug/verify/reuse), and EQY structural matching for GPT-5.5’s 3 residual FSMs.

9 Conclusion

Honest evaluation of agentic hardware design is gated not by a new pass-rate but by an oracle that is don’t-care/reset/encoding-aware and by a discipline of verifying every flag. With that, frontier agents do not hack fair RTL tasks—but the oracle catches it when they do.

Reproducibility & artifact. All code is released: the hardened oracle (`equiv.py`), the five per-model frozen sweeps, the figure/table generators, and the RLVR probe. `report_cis.py` regenerates every headline number from the frozen candidates (EDA-only, *no* LLM calls); model ids are pinned and hidden testbenches use fixed seeds, so results are deterministic (`docs/REPRODUCE.md`). The benchmark is an evaluation *environment*, so we follow the code-hosting rather than data-hosting path.

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